

Evidence I Why Each Component Is There
Every number produced by a committed command against the frozen corpus

318 English queries · graded page judgments · top_k = 10
Apple Silicon MacBook · warm retrieval, excludes model load

A · RETRIEVAL I SIX CONTROLLED RUNS, SAME SPLIT, SAME JUDGMENTS

Table with 10 columns: TEXT SOURCE, CHUNKING, RETRIEVER, NDCG@10, R@10, R@5, R@1, MRR, P50 MS, P95 MS. Rows include Shipped markdown, Shipped markdown, Shipped markdown, Shipped markdown, Shipped markdown, and Tesseract OCR.

Highlighted: the OCR arm, with a 3.0% tax against publisher text under an otherwise identical pipeline. OCR consistency over all 1,110 pages: precision 0.8564, recall 0.9461, F1 0.8990. This is a consistency signal, not a ground-truth benchmark.

B · WHERE THE TEXT PIPELINE FAILS, BY EVIDENCE TYPE

Table with 4 columns: GOLD EVIDENCE TYPE, N, NDCG@10, RECALL@10. Rows include Text, Table, Chart, and Infographic.

All 318 queries, multi-label. This routing evidence contradicts the naive expectation that charts fail. Tables carry the largest nDCG deficit; infographics the lowest recall despite the highest nDCG. The result supports targeted table recovery plus an infographic-scoped visual arm, not blanket replacement.

D · PAIRED CHUNKING COMPARISON (N=24)

Table with 5 columns: MEASURE, FIXED, STRUCTURE, Δ, W/L/T. Rows include Retrieved answer F1, Oracle control F1, Retrieved gold recall, Retrieved TTR (ms), and Noise-adjusted F1 delta.

The oracle arm receives byte-identical prompts in both runs (24/24 verified), so its movement estimates run-to-run model variance and exceeds the treatment movement. Reported as no demonstrated gain. Diagnostic, not a significance test.

C · GENERATION I RETRIEVED VS ORACLE CONTEXT

Table with 3 columns: METRIC, RETRIEVED, ORACLE. Rows include Answer token F1, Gold-page citation recall, Quote support precision, Valid context precision, Explicit refusals, Input / output tokens, TTR p50 / p95 (ms), and Cost (USD).

24 stratified query pairs, 48 live model calls, zero provider failures. Identical prompt template, schema, and token budget across both arms; only the pages entering context differ.

E · FIRST-FAILING-STAGE ATTRIBUTION

Table with 3 columns: CLASSIFICATION, RETRIEVED, ORACLE. Rows include Unscorable gold (unknown AND/OR), Retrieval incomplete, Generation failure, and Citation failure.

The unscorable majority is a deliberate refusal to over-claim, not a gap in the harness.

F · WHY EACH COMPONENT IS THERE, AND WHAT IT WAS CHOSEN OVER

Table with 3 columns: COMPONENT, WHAT IT DOES, WHY THIS, OVER THE ALTERNATIVE. Rows include ViDoRe V3 HR, Tesseract 5, BM25 (small local baseline), all-MiniLM-L6-v2, NumPy exact search, Reciprocal Rank Fusion, No LangChain / LlamaIndex, and gpt-5.4-mini.

Reproduce: pip install -e '[dev,prototype]' → dataset materialize → ocr run → index build → index dense → benchmark evaluate --mode hybrid → benchmark evidence-breakdown. Every stage carries a chained fingerprint over input artifact, implementation version, normalized config, and model revision. Changing DPI invalidates render→OCR→chunks→embeddings; changing chunk size reruns neither render nor OCR. CI runs Ruff, mypy, and the test suite. Attribution and licenses are in the README. Deliberately not built: ColPali/visual late interaction, learned reranking, native Office ingestion, a second benchmark adapter, or production deployment. Each is reasoned about rather than implemented for presentation. Not claimed: exact visual localization from a text-only path, full any-of/all-of attribution, or statistical significance from six-query evidence strata. Full decision history, including reversals, is in 09_REASONING_LOG.md (52 decisions). Corpus: 1,110 pages · 318 English queries · 1,731 English graded page judgments (~5.44 gold/query; 10,386 across all six languages) · CC-BY-4.0.